

01

The screenshot shows an IDE with a file named `Main.java`. The code defines a `Main` class with a `main` method that calls `analisar(ENTRADA)`. The `analisar` method is static and takes a `String` parameter `fonte`. It initializes `int posicao = 0;`. The output window shows the execution of the program, displaying the student's name and RA, followed by the input string `ENTRADA: total_hec = somaAnterior - valor / 02;`. The output then shows a table of tokens:

LEXEMA	TOKEN
total_hec	IDENTIFICADOR
=	ATRIBUICAO
somaAnterior	IDENTIFICADOR
-	SUBTRACAO
valor	IDENTIFICADOR
/	DIVISAO
02	INTEIRO
;	PONTO_E_VIRGULA

The process finished with exit code 0.

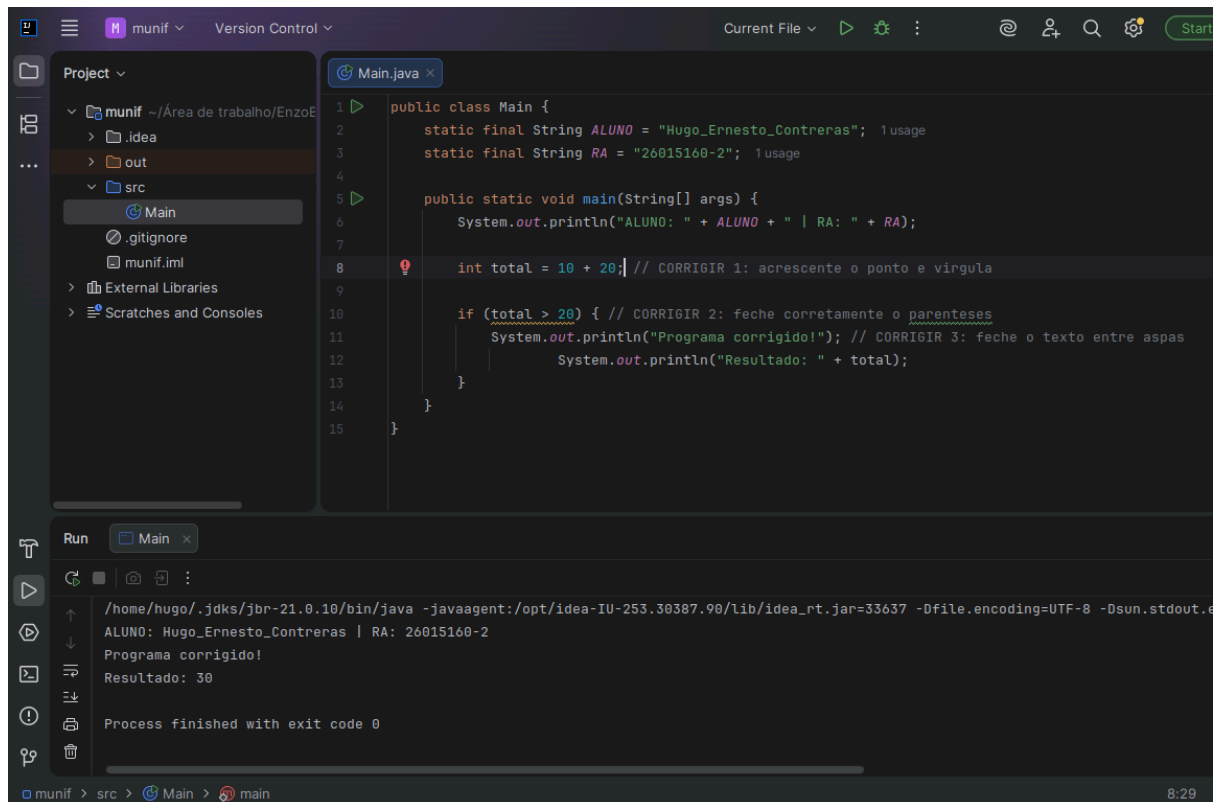
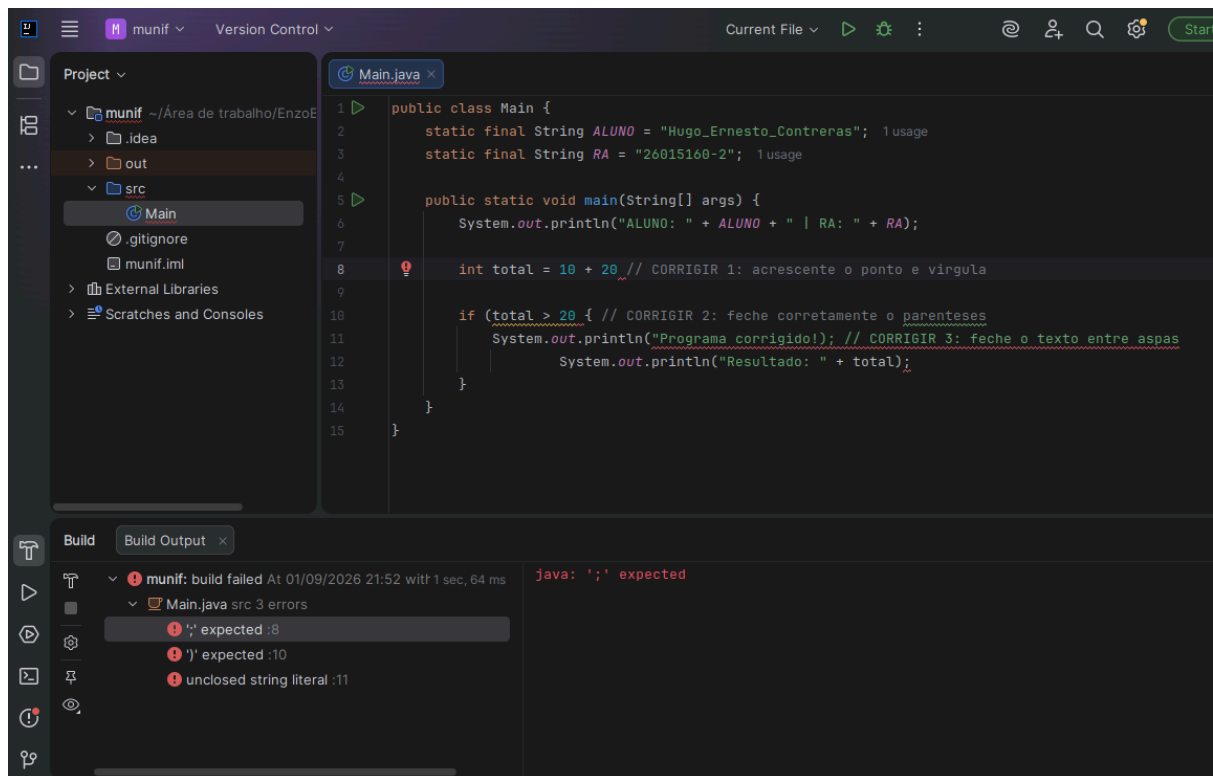
02

The screenshot shows an IDE with a file named `Main.java`. The code defines a `Main` class with a `main` method that calls `analisar(ENTRADA)`. The `analisar` method is static and takes a `String` parameter `fonte`. It initializes `int posicao = 0;`. The output window shows the execution of the program, displaying the student's name and RA, followed by the input string `ENTRADA: int nota_hec = 10;`. The output then shows a table of tokens:

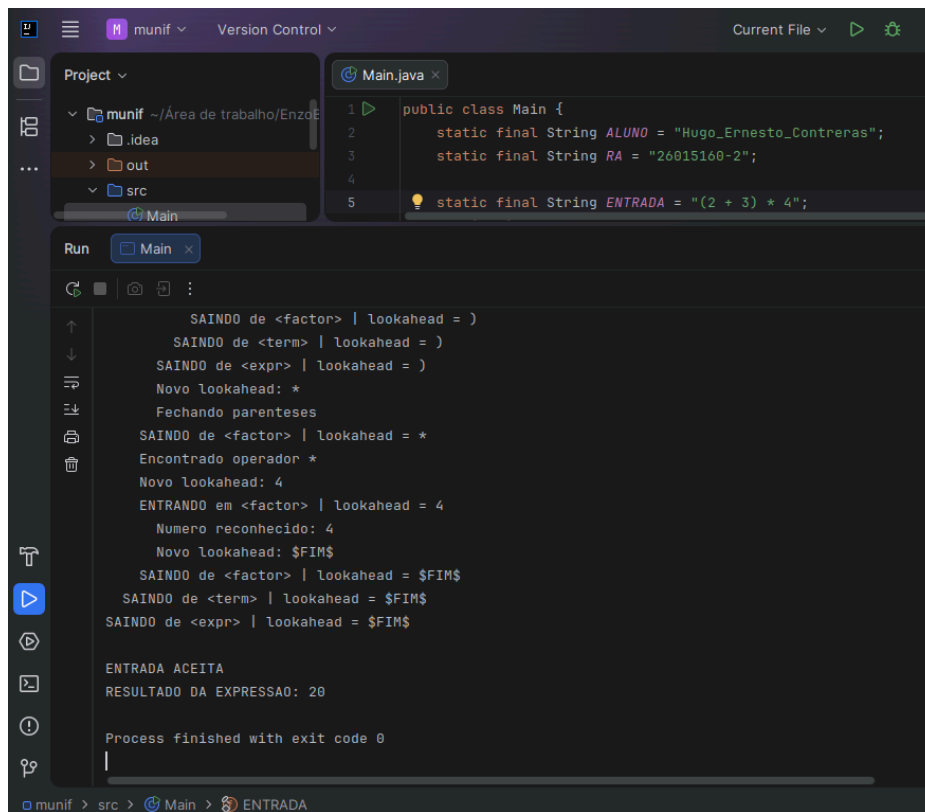
LEXEMA	CLASSE	TOKEN
int	LETRA	PALAVRA_RESERVADA
nota_hec	LETRA	IDENTIFICADOR
=	SIMBOLO	ATRIBUICAO
10	DIGITO	INTEIRO
;	SIMBOLO	PONTO_E_VIRGULA

The process finished with exit code 0.

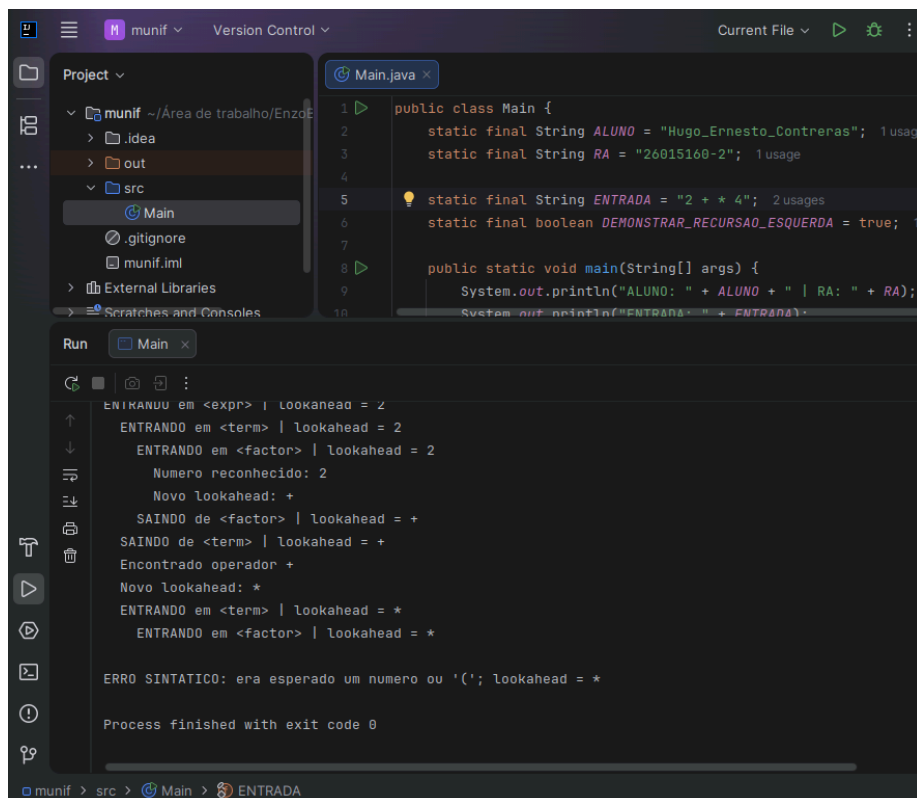
03



04



Token quando erro = '*'



A recursão a esquerda causa um loop infinito

```
/home/hugo/.jdk8/jbr-21.0.10/bin/java -javaagent:/opt/idea-IU-253.30387.90/lib/idea_rt.jar=39583 -
ALUNO: Hugo_Ernesto_Contreras | RA: 26015160-2
ENTRADA: (2 + 3) * 4

expr -> expr + term (chamada 1)
  expr -> expr + term (chamada 2)
    expr -> expr + term (chamada 3)
      expr -> expr + term (chamada 4)
        expr -> expr + term (chamada 5)
          expr -> expr + term (chamada 6)
            expr -> expr + term (chamada 7)
              expr -> expr + term (chamada 8)
                DEMONSTRACAO INTERROMPIDA: a regra chamou expr novamente sem consumir a entrada.
```

05

The screenshot shows an IDE with a project named 'munif'. The 'Main.java' file is open, showing a class with two static final strings: 'ALUNO' and 'RA'. Below the code editor, the 'Run' window displays the output of the program, which is the LR(0) item set transitions for the grammar.

Run: Main

ALUNO: Hugo_Ernesto_Contreras | RA: 26015160-2
ENTRADA: id+id*id

PILHA	ENTRADA	ACAO
\$ 0	id+id*id\$	SHIFT id -> estado 5
\$ 0 id 5	+id*id\$	REDUCE F -> id
\$ 0 F 3	+id*id\$	REDUCE T -> F
\$ 0 T 2	+id*id\$	REDUCE E -> T
\$ 0 E 1	+id*id\$	SHIFT + -> estado 6
\$ 0 E 1 + 6	id*id\$	SHIFT id -> estado 5
\$ 0 E 1 + 6 id 5	*id\$	REDUCE F -> id
\$ 0 E 1 + 6 F 3	*id\$	REDUCE T -> F
\$ 0 E 1 + 6 T 9	*id\$	SHIFT * -> estado 7
\$ 0 E 1 + 6 T 9 * 7	id\$	SHIFT id -> estado 5
\$ 0 E 1 + 6 T 9 * 7 id 5	\$	REDUCE F -> id
\$ 0 E 1 + 6 T 9 * 7 F 10	\$	REDUCE T -> T * F
\$ 0 E 1 + 6 T 9	\$	REDUCE E -> E + T
\$ 0 E 1	\$	ACCEPT

Process finished with exit code 0